

KRITIK MODI

[PORTFOLIO] [LINKEDIN] [GITHUB] [CODEFORCES] [CODECHEF] [LEETCODE] [ATCODER]

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Experience

Datatailr

Remote, New York

Founding Engineer

November 2023 - Present

- o **AI platform (retrieval, vector search, MCP)**: Lead the AI engineering work: chunking and embedding pipelines over internal corpora, a vector search layer with hybrid retrieval and reranking, and MCP servers exposing both to agents and to the products on the platform. Own model selection, evaluation, and serving.
- o **DT Query Engine (federated SQL, native C++)**: Work on the company's own federated SQL engine: ground-up C++ with LLVM 20 JIT compilation, an Apache Arrow vectorized executor, and the PostgreSQL wire protocol, so any psql/JDBC/BI client connects unchanged. 55 connectors, 340+ SQL functions, sub-second start in 100 MB where JVM engines need 2-8 GB. Ships as one standalone binary or a distributed coordinator/worker cluster, joining 100K+ tables across sources with zero ETL.
- o **Multi-cloud enablement (AWS, Azure, GCP, Hetzner)**: Defined the product itself as code in Terraform across four clouds (compute, storage, networking, containers, IAM), with Python and C++ orchestration layers, automated SQL provisioning, one-click installs, and GitHub Actions for testing and provisioning. Enabled 10+ business deployments, driving +250% ARR / \$500k pipeline.
- o **Private Networking Architecture**: Built a private network cluster on Azure (VNETs, subnets, firewalls, private DNS zones, Private Endpoints) for internet-isolated service access, unlocking a security-heavy revenue model.
- o **Observability and cost optimizations**: Instrumented cron and Nomad metrics through Statsite behind an NGINX proxy, then sized jobs from historical usage via a right-sizing algorithm and backend APIs, reducing costs by ~75%.
- o **Batch orchestration & graph tooling**: Defined DAGs for batch pipelines and built a dependency visualizer that orders stages via topological sort, with node-level observability and retries in a custom UI.
- o **Client Engineering**: Worked directly with enterprise clients and prospects to scale production use cases end to end, from debugging critical issues to surfacing product gaps and driving fixes.

Juspay

Hybrid, Bangalore

Software Development Engineer

April 2023 - November 2023

- o **Payment orchestrator (Hyperswitch, open-source)**: Contributed Rust features/fixes to a payment routing engine (idempotency keys, retry/backoff, input validation) for safe, efficient processing under high concurrency.
- o **Backend SDKs (Hyperswitch integrations)**: Designed and published multiple backend SDKs for Hyperswitch integrations across stacks (Node.js, PHP, Go, Ruby, Java), with JavaScript/HTML/CSS frontend components; open-sourced for external adoption. [\[LINK\]](#).
- o **Release engineering & DevOps**: Dockerized services and authored Jenkins pipelines to build/test/publish artifacts; managed AWS configs and rollouts; deployed Kubernetes pods for backup.

Achievements

- **Smart India Hackathon 2022**: Winner · Built a scalable fake news detection system for Twitter, flagging misinformation from spam accounts via multi-parameter analysis · Awarded by the Government of India. [\[LINK\]](#)
- **GitHub Externship 2022**: Selected as a GitHub Extern among 14000+ applicants, a selection rate of 0.6%. [\[LINK\]](#)
- **ACM-ICPC 2022**: Regionalist; qualified for the Kanpur-Mathura regionals and ranked amongst the top 80 teams onsite.
- **Meta Hacker Cup 2024**: Achieved India Rank 186 and 85 in Hacker Cup Round 1 and Round 2 respectively.
- **Google Code Jam 2023**: Achieved India Rank 248 and 43 in Code Jam Farewell Round A and Round C respectively.
- **LeetCode (Guardian)**: Peak rating: 2488 · Top 0.2% global, Top 0.03% India · Best rank: 32 among 30000+ contestants.
- **CodeChef (6*)**: Peak rating: 2222 · Top 0.2% global, Top 0.05% India · Best rank: 11 among 3000+ contestants.
- **Codeforces (Expert)**: Peak rating: 1790 · Top 0.5% global · Best rank: 23 among 14000+ contestants in Round 1006.
- **Research Publication**: Published a literature survey on misinformation flagging systems, covering text classification, feature extraction, and network-based signals. [\[LINK\]](#)

Skills Summary

- **Languages**: Java, Python, C++, Rust, JavaScript, SQL, Bash
- **Infrastructure & Cloud**: Terraform, AWS, Azure, GCP, Kubernetes, Docker, Nomad, Hetzner, Jenkins, GitHub Actions
- **Data & Platforms**: PostgreSQL, MySQL, MongoDB, Databricks, Hadoop/Hive, Snowflake, BigQuery, Redshift, Kafka, S3

Education

Dayananda Sagar College of Engineering

B.E. Computer Science and Engineering, CGPA: 9.10

Bangalore, India

2019 - 2023

Coding Profiles

- **Competitive Programming**: 250+ rated contests, 2000+ problems solved: [\[Codeforces\]](#) [\[CodeChef\]](#) [\[LeetCode\]](#) [\[AtCoder\]](#)
- **GitHub**: Open-source and projects: [\[github.com/kritikmodi\]](#) · **Selected repos**: [\[Codeware\]](#) [\[CollegeLink\]](#) [\[Portfolio\]](#)